TANGGAL PERCOBAAN : Jumat, 09-10-2026

TANGGAL PENGUMPULAN : Jum'at, 16-10-2026

PRAKTIKUM PEMROGRAMAN KOMPUTER SEMESTER 125

MODUL – 3 : TRIGONOMETRI

Mencerdaskan & Memartabatkan Bangsa

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Laporan Awal	Laporan Akhir	Total
88		

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2026

Dipindai dengan CamScanner

Modul - 3
Trigonometri

1) Problem Statement

Membuat Program Untuk Menghitung Nilai Sin dan Cos dengan Perhitungan deret Mclaurin

2) Mathematical equation

a. Deret Mclaurin Sinus = $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

b. Deret Mclaurin Cosinus = $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

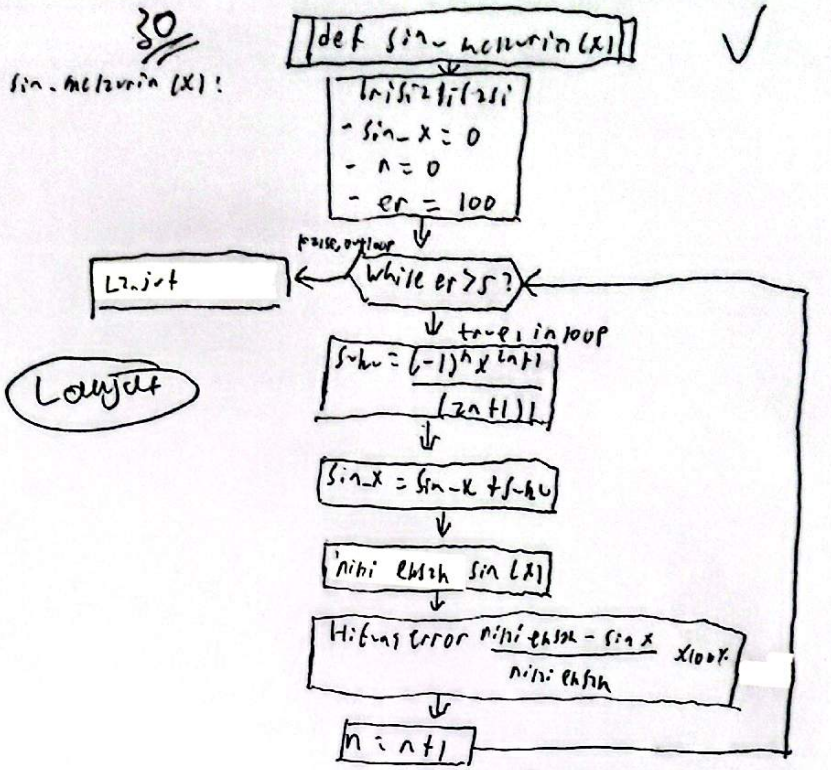
3) Algoritma

1. Mulai
2. import math
3. Print identitas : - nama
   - nim
4. def sin_mclaurin (x):
   4.1 inisialisasi
   - sin_x = 0
   - n = 0
   - er = 100
   4.2 While er >= 5
   4.2.1. hitung suku ke n dari deret mclaurin : suku = $\frac{(-1)^n \cdot x^{2n+1}}{(2n+1)!}$
   4.2.2. tambahkan suku ke sin x
   4.2.3. Hitung nilai eksak sin(x)
   4.2.4. Hitung error : er = abs (nilai_eksak - sin_x)/nilai eksak x100 %
   4.2.5. n + 1
   4.3. kembalikan sin_x dan er
5. def fungsi cos_mclaurin (x):
   5.1. inisialisasi
   - cos_x = 0
   - n = 0
   - er = 100
   5.2. while er >= 5.
   5.2.1. Hitung suku ke n dari deret mclaurin : suku = $\frac{(-1)^n \cdot x^{2n}}{(2n)!}$
   5.2.2. Tambahkan suku ke cos_x
   5.2.3. Hitung nilai eksak cos(x)
   5.2.4. Hitung error : er = abs ((nilai_eksak - cos x)/nilai eksak) x100 %
   5.2.5. n + 1
   5.3. kembalikan cos_x dan er
6. input float "derajat"
7. konversi derajat ke radian
8. panggil fungsi sin mclaurin dan cos mclaurin
9. print "nilai sin(x) & error sin | nilai cos(x) & error cos"
10. selesai

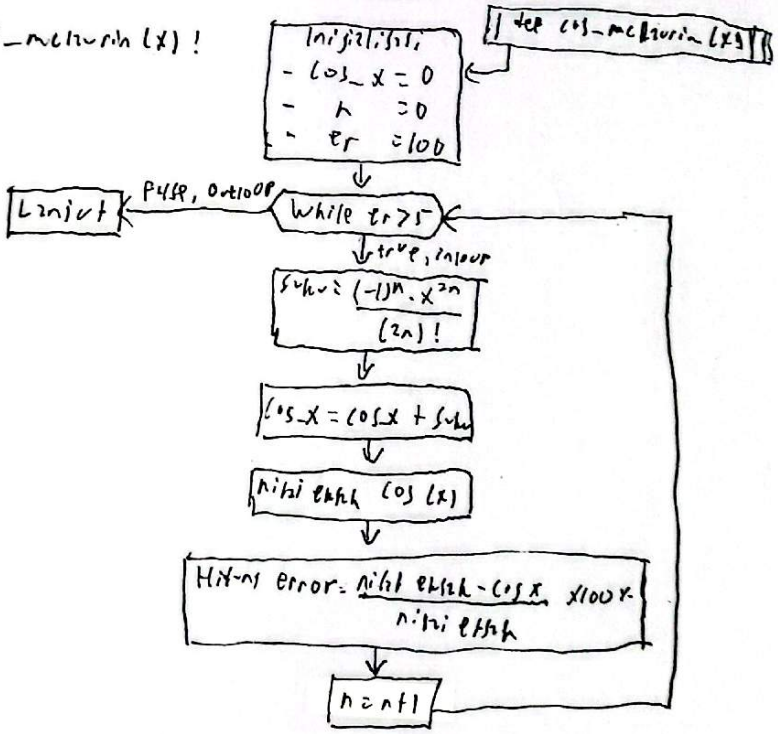
Kebawah ↓

4) Flowchart 30

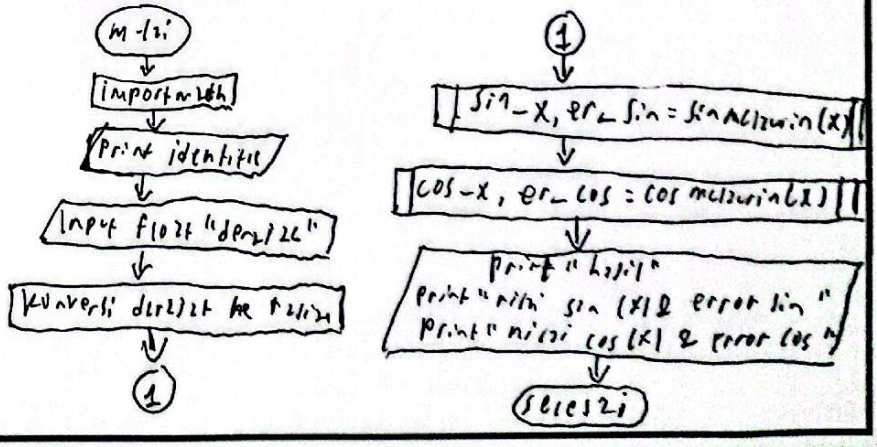
2. Flowchart def sin_maclaurin(x):



b. Flowchart def cos_maclaurin(x):



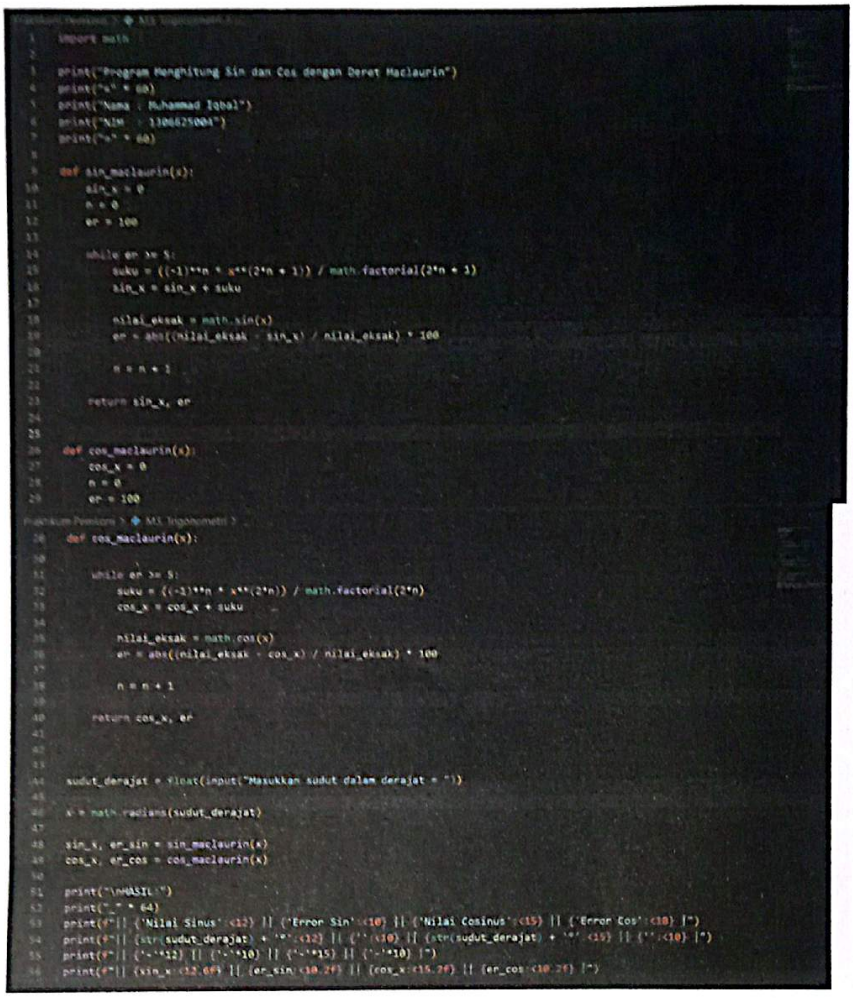
c. Main Flowchart = ∫



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Input&Output-SC-Modul-2-1306625004-MI

Source code:



```
import math

print("Program Menghitung Sin dan Cos dengan Deret Maclaurin")
print("=" * 60)
print("Nama : Muhammad Iqbal")
print("NIM  : 1306625004")
print("=" * 60)

def sin_maclaurin(x):
    sin_x = 0
    n = 0
    er = 100

    while er >= 5:
        suku = ((-1)**n * x**(2*n + 1)) / math.factorial(2*n + 1)
        sin_x = sin_x + suku

        nilai_eksak = math.sin(x)
        er = abs((nilai_eksak - sin_x) / nilai_eksak) * 100

        n = n + 1

    return sin_x, er


def cos_maclaurin(x):
    cos_x = 0
    n = 0
    er = 100
Praktikum Pemkom > M3_Trigonometri >
def cos_maclaurin(x):

    while er >= 5:
        suku = ((-1)**n * x**(2*n)) / math.factorial(2*n)
        cos_x = cos_x + suku

        nilai_eksak = math.cos(x)
        er = abs((nilai_eksak - cos_x) / nilai_eksak) * 100

        n = n + 1

    return cos_x, er

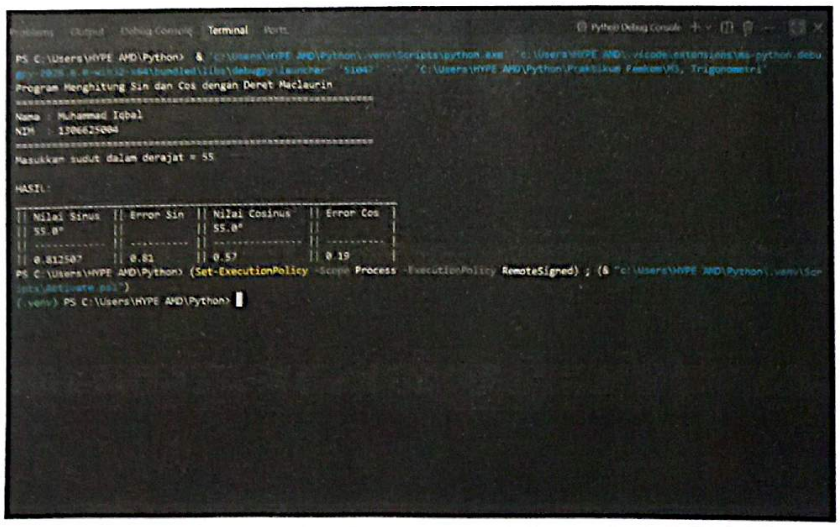

sudut_derajat = float(input("Masukkan sudut dalam derajat = "))

x = math.radians(sudut_derajat)

sin_x, er_sin = sin_maclaurin(x)
cos_x, er_cos = cos_maclaurin(x)

print("\nHASIL:")
print("_" * 64)
print(f"| {'Nilai Sinus':<12} | {'Error Sin':<10} | {'Nilai Cosinus':<15} | {'Error Cos':<10} |")
print(f"| {str(sudut_derajat) + '°':<12} | {'':<10} | {str(sudut_derajat) + '°':<15} | {'':<10} |")
print(f"| {'-'*12} | {'-'*10} | {'-'*15} | {'-'*10} |")
print(f"| {sin_x:<12.6f} | {er_sin:<10.2f} | {cos_x:<15.2f} | {er_cos:<10.2f} |")
```

Output:



```
Problems  Output  Debug Console  Terminal  Ports                    Python Debug Console

PS C:\Users\HYPE AMD\Python> & 'c:\Users\HYPE AMD\Python\.venv\Scripts\python.exe' 'c:\Users\HYPE AMD\.vscode\extensions\ms-python.debu
gpy-2025.8.0-win32-x64\bundled\libs\debugpy\launcher' '51047' '--' 'C:\Users\HYPE AMD\Python\Praktikum Pemkom\M3, Trigonometri'
Program Menghitung Sin dan Cos dengan Deret Maclaurin
============================================================
Nama : Muhammad Iqbal
NIM  : 1306625004
============================================================
Masukkan sudut dalam derajat = 55

HASIL:
________________________________________________________________
| Nilai Sinus  | Error Sin  | Nilai Cosinus   | Error Cos  |
| 55.0°        |            | 55.0°           |            |
| ------------ | ---------- | --------------- | ---------- |
| 0.812507     | 0.01       | 0.57            | 0.19       |
PS C:\Users\HYPE AMD\Python> Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned ; & "c:\Users\HYPE AMD\Python\.venv\Scr
ipts\Activate.ps1"
(.venv) PS C:\Users\HYPE AMD\Python>
```

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